VOICE ARENA RESEARCH LAB

Speaker Diarization Leaderboard

Production-grounded, multi-domain benchmark ranking the accuracy, cross-talk robustness, latency, and cost of leading speech diarization engines.

120.4 hrs	10	0ms Collar
AUDIO EVALUATED	ACTIVE MODELS	PRIMARY METRIC

DOMAIN: All Domains | Enterprise Meetings | Telephony (8kHz) | Clinical Consults | Noisy Public

TRACK: Batch (Offline) | Streaming (<500ms)

Boundary Collar: 0ms (Strict) | 250ms (Legacy)

RANK	MODEL & PROVIDER	CATEGORY	PRIMARY DER (0MS) ± 95% CI	OVERLAP DER	JER (FAIRNESS)
#1	PA pyannote 3.1 pyannote.audio • Self-Hosted	Open-Source	11.4% ± 0.5%	22.8%	15.1%
#2	DG Nova-2 (Diarize) Deepgram • Managed Cloud API	Joint ASR+Diarize	12.1% ± 0.6%	24.5%	16.3%

RANK	MODEL & PROVIDER	CATEGORY	PRIMARY DER (0MS) ± 95% CI	OVERLAP DER	JER (FAIRNESS)
#3	NM NeMo (TitaNet + MSDD) NVIDIA • Self-Hosted	Open-Source	12.7% ± 0.7%	25.1%	17.0%
#4	AA Universal-2 (Speaker Labels) AssemblyAI • Managed Cloud API	Joint ASR+Diarize	13.5% ± 0.6%	27.9%	18.2%
#5	WX WhisperX v3.1 m-bain / Open Source • Self-Hosted	Open-Source	13.9% ± 0.8%	28.4%	18.9%
#6	AZ Azure Speech Diarization Microsoft • Cloud API / Container	Cloud Enterprise	14.6% ± 0.7%	31.2%	19.8%
#7	RV Rev Diarization API Rev.com • Managed Cloud API	Cloud Enterprise	15.2% ± 0.8%	32.6%	20.4%

RANK	MODEL & PROVIDER	CATEGORY	PRIMARY DER (0MS) ± 95% CI	OVERLAP DER	JER (FAIRNESS)
#8	**Chirp STT V2 Diarizer** GC Google Cloud • Managed Cloud API	Joint ASR+Diarize	15.8% ± 0.9%	34.1%	21.5%
#9	**AWS Transcribe Multi-Speaker** AW Amazon AWS • Managed Cloud API	Joint ASR+Diarize	17.4% ± 0.8%	38.5%	23.2%
#10	**SpeechBrain ECAPA-TDNN** SB SpeechBrain • Self-Hosted	Open-Source	19.2% ± 1.1%	46.2%	26.8%

⚖️ Why 0ms Boundary Collar?

Traditional benchmarks use a 250ms collar that forgives 500ms of boundary uncertainty per turn. Modern voice bots cannot tolerate cross-talk hallucinations, so Voice Arena scores at true acoustic onset/offset.

👥 Unknown Speaker Count

No model is provided the number of speakers upfront. Systems must autonomously cluster x-vectors and discover the speaker count, mirroring production reality.

📊 Bootstrap Significance

All scores are computed with 1,000 bootstrap iterations. Pairwise rankings are confirmed with Wilcoxon signed-rank tests ($p < 0.01$) to eliminate ranking by chance.